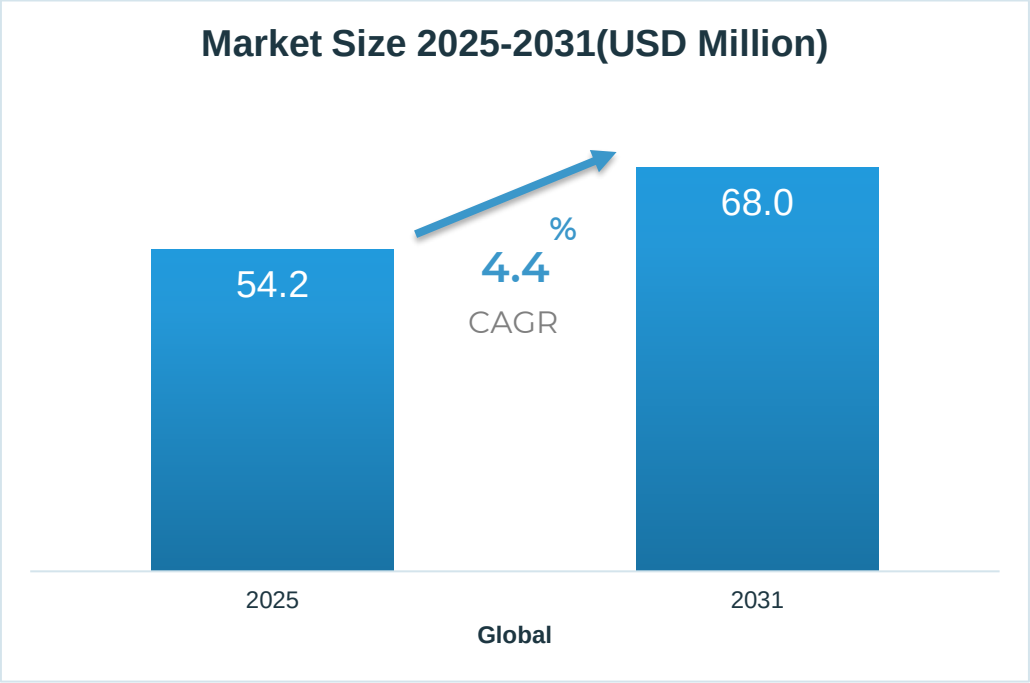
MILITARY AIRCRAFT MODERNIZATION MARKET – SIZE, SHARE, TRENDS, GROWTH FORECAST, AND COMPETITIVE ANALYSIS (2025-2030)

TITLE**:** The Military Aircraft Modernization Market is segmented **by Aircraft Type** (Fighter jets, Transport Aircraft, Attack Helicopters, Transport/Utility Helicopters, Bombers, Special Mission Aircraft), **by Customer Type** (U.S Air force, Navy/Marine Corps, Army Aviation, Europe (NATO Allies), Asia Pacific Allies, Middle East FMS, Latin America & ROW, Government Depots In-House, Commercial Contractors), **by Contract Value Band** (Mega Programs, Large Medium , Small), **by Supply Chain Tier** (Tier 1,Tier 2,Tier 3,Tier 4), **by Platform Age** (50+ Years Old, 30-49 Years Old, 20-29 Years Old, 10-19 Years Old, <10 Years Old), **by Geography** (North America, Europe, Asia-Pacific, Middle East, Africa and Latin America).

FIGURE 1 MARKET SIZE 2025-2031(USD MILLION)



REPORT OVERVIEW

The global military aircraft modernization market has become essential of air superiority programs, due to rising geopolitical tensions and the need for sustained combat aircraft capability enhancement. In **2024**, the market is estimated at **USD 52 billion** and is expected to reach **USD 68.04 billion** by **2030**, supported by extensive fighter jet upgrade activity, growth in the avionics upgrade market, and widespread defense aircraft lifecycle extension initiatives. The market is projected to grow at an estimated **4-5.5 %** CAGR, as governments favor military aircraft modernization over full fleet replacement to optimize defense spending allocation.

FIGURE 2 MARKET DYNAMICS

DRIVERS

- **Air superiority and combat readiness focus**
  Increasing emphasis on **combat aircraft capability enhancement** to counter advanced air defenses, hypersonic threats, and electronic warfare is accelerating **military aircraft modernization** programs.

- **Cost-effective alternative to new procurement**
  Military jet refurbishment and **defense aircraft lifecycle extension** enable faster deployment and lower capital intensity than new aircraft acquisition, supporting efficient **fighter jet upgrade** strategies.

- **Growth of military aviation technology**
  Expansion of the **avionics upgrade market**, including AESA radars, electronic warfare systems, and network-centric mission computers, is significantly enhancing **legacy aircraft enhancement**.

- **Stealth and survivability integration**
  Rising adoption of **stealth technology retrofit**, low-observable coatings, and electronic countermeasures is improving survivability of non-stealth platforms in contested airspace.

- **Long-term military procurement alignment**
  Multi-decade **military procurement programs** increasingly balance **aging fleet replacement**, **tactical aircraft modernization**, and **next generation fighter** development.

RESTRAINTS

- **Pressures on defense spending allocation**
  Changes in **defense spending allocation** patterns between air, naval, cyber, and space segments make it difficult to allocate funds for comprehensive **military aircraft modernization**.

- **High capital intensity of advanced upgrades**
  The capital costs of **stealth technology upgrade**, advanced **radar system upgrade**, and **fifth-generation avionics** limit the upgrade scope, especially in mid-life **legacy aircraft upgrade** programs.

- **Dependence on long procurement cycles**
  The long approval, contracting, and funding cycles in **military procurement programs** constrain market reaction to the rapidly changing threat environment.

- **Platform obsolescence risk**
  Some aging platforms face diminishing upgrade viability, capping returns on **defense aircraft lifecycle extension** and accelerating selective **aging fleet replacement** decisions.

OPPORTUNITIES

- **F-35 and fifth-generation upgrade roadmaps**
  F-35 upgrade and **fifth-generation fighter upgrade** programs drive a continuous demand for software-defined avionics, sensors, and **weapons integration**.

- **Growing demand from new air forces**
  The Asia-Pacific, Middle East, and Eastern European regions are moving forward with **tactical aircraft upgrades** to fill existing gaps without necessarily replacing their entire **aging aircraft fleet**.

- **Open-architecture upgrades**
  The use of open mission systems makes it easier to upgrade the **defense aircraft lifecycle**.

- **Defense contractor and subsystem expansion**
  F-35 upgrade and **fifth-generation fighter upgrade** programs drive a continuous demand for software-defined avionics, sensors, and **weapons integration**.

CHALLENGES

- **Complexity of aircraft weapon system integration**
  The integration of **aircraft weapon systems** from multiple vendors is a complex task that poses risks to technical integration, interoperability, and timelines.

- **Synchronization with future fighter programs**
  Synchronizing the modernization of **fighter aircraft upgrades** with the introduction of **next-generation fighters** and **fifth-generation fighters** is an operational challenge.

- **Certification, testing, and airworthiness burden**
  The burden of flight testing and certification creates delays in the introduction of new **military aviation technology**.

- **Cyber security and data-fusion execution risk**
  The increasing digitization of combat aircraft makes data fusion a challenge to the real-time enhancement of **combat aircraft capabilities**.

FIGURE 3 MARKET ANALYSIS DASHBOARD

KEY HIGHLIGHTS

- The military aircraft modernization industry was valued at **USD 52 billion in 2024** and is projected to reach **USD 68.0 billion by 2030**, growing at a **4.6% CAGR**, driven mainly by modernization-led procurements rather than new aircraft acquisitions.
- By geography**, North America** leads with **50% market share in 2024**, due to ongoing U.S. fighter and bomber upgrade programs, while **Asia-Pacific** is the fastest-growing region at **8.2% CAGR** from 2024 to 2030.
- By aircraft type, **fighter aircraft** account for **45%** of total market value in 2024 and are expected to reach **USD 30.8 billion** by 2030.
- By customer type**,** the **U.S. Air Force** represents **36%** of market share and grows at a **5.4% CAGR**, highlighting sustained investment in avionics, radar, and weapon system upgrades.
- By contract value band, **mega** modernization programs contribute **35%** of total contract value in 2024, reflecting the market's concentration toward large, long-term, platform-scale modernization contracts.
- By supply chain tier**, Tier-1** prime contractors hold **44%** market share and grow at a **5.1% CAGR**, emphasizing OEM-led system integration and lifecycle support dominance.
- By platform age**, aircraft aged 30–49 years** account for **40%** of modernization spending, while **50+ year-old platforms** record the fastest growth at **5.6% CAGR**, confirming lifecycle extension as the primary strategy amid delayed next-generation fighter induction.

Regional Insights

- **North America dominated the Military Aircraft Modernization Market with the highest revenue share in 2024 and is expected to maintain its leadership position through 2030.** North America alone accounted around 50% market share in the year 2024, driven by sustained U.S. defense expenditure, large-scale fleet upgrade programs for fighter jets, transport aircraft, and ISR platforms, and continuous investments in advanced avionics, electronic warfare systems, and next-generation radar technologies. The presence of leading defense contractors, strong government funding for modernization of legacy aircraft, and strategic focus on maintaining air superiority further strengthen North America's market dominance. Additionally, increasing emphasis on multi-domain operations, AI-enabled combat systems, and interoperability within NATO allies continues to accelerate modernization initiatives across the region.

FIGURE 4 MILITARY AIRCRAFT MODERNIZATION MARKET TRENDS BY REGION 2024-2030

COMPANY HIGHLIGHTS:

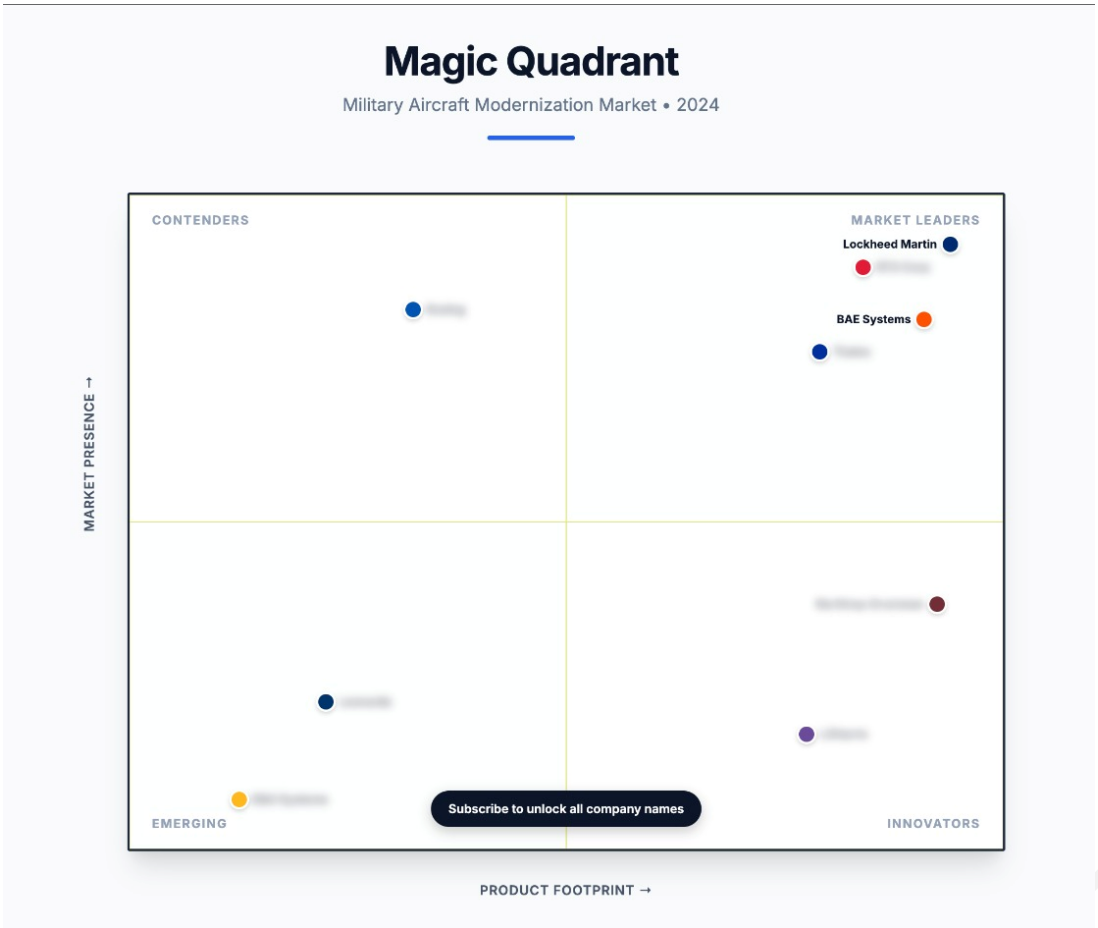
- The U.K. Ministry of Defence has shortlisted **Lockheed Martin, BAE Systems, and Thales Group** for its **Project NYX** autonomous support drone programme related to Apache helicopters, reflecting an expanding modernisation agenda in unmanned and support aircraft.
- **Thales** has recently awarded contracts for **RBE2 AESA** radar structures, supporting modernised fighter avionics and sensor suites for export customers and reinforcing its role in next-generation radar supply chains.

Military Modernization Market Companies

Following are the leading companies in Military Aircraft Modernization Market. These companies collectively hold the largest market share and dictate the industry trends

1. LOCKHEED MARTIN
2. THALES GROUP
3. RTX CORPORATION
4. GENERAL DYNAMICS
5. BAE SYSTEMS
6. BOEING
7. NORTHROP GRUMMAN
8. L3HARRIS TECH
9. LEONARDO S.P.A.
10. ELBIT SYSTEMS

FIGURE 5 MAGIC QUADRANT MILITARY MODERNIZATION MARKET 2024



RECENT DEVELOPMENTS

1. Lockheed Martin

★ **Record $176 Billion Backlog & $71 Billion Revenue — World's Largest Defense Contractor (2024)**

Lockheed Martin achieved a record year-end backlog of $176 billion in 2024 — the largest contract backlog in the company's history — alongside net sales of $71.0 billion (5% YoY growth), cementing its position as the world's largest defense contractor by revenue. The F-35 Lightning II program, the most expensive weapons system ever built at $400B+ lifecycle cost, remains the company's crown jewel, with 3,000+ aircraft on order across 17 nations. Lockheed invested over $3 billion in R&D and capital expenditure in 2024 to advance its 21st Century Security vision, while returning $6.8 billion to shareholders through dividends and buybacks.

Source: *Lockheed Martin Q4 & Full Year 2024 Earnings Press Release, January 28, 2025*

URL: *https://news.lockheedmartin.com/2025-01-28-Lockheed-Martin-Reports-Fourth-Quarter-and-Full-Year-2024-Financial-Results*

2. Thales Group

★ **Record €51 Billion Order Book & 11.7% Revenue Growth to €20.6 Billion in 2024**

Thales Group delivered its strongest financial performance in recent history in 2024: revenue surged 11.7% to €20,577 million, with an order book reaching an all-time high of nearly €51 billion — a €5.4 billion increase year-over-year. Adjusted EBIT reached €2,419 million (11.8% margin) and net income surged 39% to €1,420 million. The year also marked Thales's successful $3.6 billion acquisition of Imperva (cybersecurity leader) and the €2.5 billion divestiture of its ground transportation division to Hitachi Rail — a strategic repositioning toward Defence, Cybersecurity, and Digital Identity that positions Thales as Europe's premier dual-domain defense and security technology champion.

Source: *Thales Group 2024 Full-Year Results Press Release, March 2025*

URL: *https://www.thalesgroup.com/en/news-centre/press-releases/thales-reports-its-2024-full-year-results*

3. RTX Corporation

★ **$80.8 Billion Adjusted Sales — 11% Organic Growth & Patriot Missile System Global Dominance (2024)**

RTX Corporation achieved adjusted sales of $80.8 billion in 2024, representing 11% organic sales growth and 13% adjusted EPS growth — the strongest performance since its formation through the merger of Raytheon and United Technologies in 2020. Raytheon's Patriot missile defense system secured over $3 billion in new contracts across 2023–2024, covering Poland, Romania, and multiple undisclosed NATO allies, making RTX the backbone of Allied air and missile defense globally. The Pratt & Whitney F135 engine — which powers 100% of F-35 aircraft — and GTF commercial engine family collectively represent the world's most widely deployed next-generation military and commercial propulsion system.

Source: *RTX Q4 & Full Year 2024 Results, January 28, 2025*

URL: *https://investors.rtx.com/static-files/ceebbf85-eb69-4563-a303-c62ad9918fba*

4. General Dynamics

★ **$47.7 Billion Revenue & Virginia-Class Submarine Program — Critical Pillar of U.S. Naval Superiority (2024)**

General Dynamics generated $47.7 billion in revenue in 2024, driven by the Electric Boat division's Virginia-Class submarine program — the U.S. Navy's primary attack submarine and a cornerstone of American undersea dominance. Electric Boat is also the prime contractor for the Columbia-class ballistic missile submarine, the most critical nuclear deterrence platform in the U.S. arsenal, with a total program value exceeding $100 billion. General Dynamics' Gulfstream business aviation segment delivered record revenues, and the Combat Systems division secured major contracts for Abrams tank upgrades and Stryker armored vehicles, reinforcing GD's role as the primary supplier across all three domains of U.S. military superiority: sea, land, and air.

Source: *General Dynamics 2024 Annual Report / Investor Relations*

URL: *https://www.gd.com/investor-relations*

5. BAE Systems

★

Record £25.3 Billion Revenue — AUKUS Nuclear Submarine Programme & European Defense Surge (2024)

BAE Systems delivered record revenue of £25.3 billion in 2024 (+14% YoY), driven by surging European defense budgets, the AUKUS nuclear-powered submarine programme, and the F-35 production partnership with Lockheed Martin. BAE Systems is the principal industrial partner for the SSN-AUKUS submarine program — a landmark trilateral partnership between the UK, Australia, and the US valued at over $360 billion over its lifetime — and the world's largest producer of naval vessels and combat aircraft after the U.S. primes. The order backlog exceeded £74 billion, reflecting the strongest demand environment in BAE's modern history driven by the post-Ukraine European rearmament wave.

Source: *BAE Systems 2024 Annual Results Press Release*

URL: *https://www.baesystems.com/en/our-company/investor-relations/results-and-reports*

6. Boeing

★

T-7A Red Hawk First Production Delivery & $22+ Billion Defense Backlog Sustained Under New Leadership (2024)

Boeing Defense delivered the first production T-7A Red Hawk advanced jet trainer to the U.S. Air Force in 2023 — a landmark achievement for a program developed with 80% digital engineering, cutting development time by 75% compared to traditional methods. Despite commercial challenges, Boeing's defense backlog exceeded $60 billion in 2024, with the MQ-25 Stingray (world's first carrier-based unmanned aerial refueling tanker), KC-46 Pegasus tanker, and F-15EX Eagle II programs anchoring long-term DoD relationships. New CEO Kelly Ortberg, appointed August 2024, initiated a comprehensive stabilization plan to restore Boeing's operational excellence and delivery performance.

Source: *Boeing 2024 Investor Relations / Defense Programs Update*

URL: *https://investors.boeing.com*

7. Northrop Grumman

★ **B-21 Raider First Flight & $84.2 Billion Backlog — America's Next-Generation Nuclear Stealth Bomber (2023–2024)**

Northrop Grumman completed the historic first flight of the B-21 Raider — the world's first sixth-generation stealth bomber — in November 2022, with flight testing continuing through 2024 toward operational deployment. The B-21 is the U.S. Air Force's primary nuclear deterrence platform of the 21st century, with a program value projected at $200+ billion over its lifecycle. Northrop's Space Systems segment achieved $13.9 billion in revenue in 2023 (+14% YoY), and the company's total backlog reached $84.2 billion — reflecting its position as the prime contractor for America's most strategically critical defense programs including Ground Based Strategic Deterrent (GBSD) intercontinental ballistic missile replacement.

Source: *Northrop Grumman 2023 Annual Report / SGCR Defense Analysis*

URL: *https://selectgcr.com/government-contracts/10-best-contractors-military-companies-by-revenue/*

8. L3Harris Technologies

★ **$21+ Billion Revenue & $3.5 Billion SDA Missile-Tracking Satellite Contracts — Space & Electronic Warfare Leader**

L3Harris Technologies generated $21+ billion in annual revenue and secured a position on every tranche of the Space Development Agency's Proliferated Warfighter Space Architecture (PWSA) missile tracking constellation — including a $843 million Tranche 3 contract in December 2025 — making it the only company to participate in all PWSA tracking layer tranches from Tranche 1 through Tranche 3. L3Harris also pioneered the DiSCO (Distributed Spectrum Collaboration and Operations) electronic warfare capability and modernized F/A-18 electronic warfare systems, reinforcing its position as the U.S. military's primary tactical communications and electronic warfare systems provider across air, land, sea, and space domains.

Source: *SDA Tranche 3 Award / Breaking Defense / DefenseScoop, December 2025*

URL: *https://defensescoop.com/2025/12/19/sda-tranche-3-missile-tracking-layer-contract-awards/*

9. Leonardo S.p.A.

★ **Record €17.8 Billion Revenue & Eurofighter Typhoon + AW149 Helicopter Global Expansion (2024)**

Leonardo S.p.A. achieved record revenues of €17.8 billion in 2024, driven by strong performance across its helicopters, defense electronics, and aerostructures divisions. Leonardo is the global leader in naval helicopter manufacturing, supplying the AW101, AW149, and AW169 across 30+ nations — with new contracts for the AW149 across Poland, South Korea, and Middle Eastern customers representing billions in export value. As a key Eurofighter Typhoon partner alongside BAE Systems, Airbus, and MBDA, Leonardo's share in the program covers avionics, radar systems, and fuselage components for 600+ delivered aircraft across 8 nations, positioning Leonardo as Europe's most internationally diversified defense electronics and aerospace company.

Source: *Leonardo S.p.A. 2024 Annual Results / Investor Relations*

URL: *https://www.leonardocompany.com/en/investors/financial-reporting*

10. Elbit Systems

★ **$1.635 Billion Single European Contract & Record $6.5 Billion Annual Revenue — Global Defense Export Champion**

Elbit Systems was awarded a landmark $1.635 billion contract from a European country in 2024 — one of the largest single defense export contracts ever awarded to an Israeli defense company — covering long-range precision strike artillery-rocket systems, unmanned aerial combat systems, and AI-embedded reconnaissance platforms. Elbit achieved record annual revenues of approximately $6.5 billion in 2024, with an order backlog exceeding $22 billion — driven by unprecedented global demand following Russia's invasion of Ukraine. Elbit is the world's largest supplier of helmet-mounted display systems, active protection systems, and tactical unmanned aerial systems, with 85% of revenues generated from international customers across 100+ countries.

Source: *Elbit Systems Press Release 2024 / Investor Relations*

URL: *https://www.elbitsystems.com/news/elbit-systems-awarded-1635-billion-contract-deliver-range-defense-solutions-european-country*

MILITARY MODERNIZATION MARKET 2024-2030 REPORT SCOPE

Report Attribute	Details
Market size value in 2024	USD 52.0 Billion
Revenue forecast in 2030	USD 68.04 Billion
Growth rate	CAGR of 4.6% from 2025 to 2030
Base year for estimation	2024
Forecast period	2025 – 2030
Quantitative units	Revenue in USD Million/Billion and CAGR from 2025 to 2030
Report coverage	Revenue forecast, market share analysis, competitive landscape, contract value assessment, supply chain mapping, modernization trends, platform lifecycle analysis, regional outlook, and strategic benchmarking
Segments covered	Aircraft Type, Customer Type, Contract Value Band, Supply Chain Tier, Platform Age, Geography
Regional scope	North America; Europe; Asia-Pacific; Middle East; Africa; Latin America
Country scope	U.S.; Canada; UK; Germany; France; Italy; Spain; Poland; India; China; Japan; South Korea; Australia; Saudi Arabia; UAE; Israel; Turkey; Brazil; Mexico; South Africa
Key companies profiled	Lockheed Martin; Boeing Defense; Northrop Grumman; BAE Systems; Raytheon Technologies; Thales Group; Airbus Defence & Space; Leonardo S.p.A.; Saab AB; Elbit Systems
Customization scope	Free report customization (equivalent up to 8 analysts working days) with purchase. Addition or alteration to country, regional & segment scope available.
Pricing and purchase options	Flexible licensing options available (Single User, Multi-User, Enterprise). Customized purchase options available based on specific modernization segments or regions.

Military Aircraft Modernization Report Market Segmentation

Global Military Aircraft Modernization Market Segmentation Analysis

This report forecasts revenue growth at global, region, and country levels and provides an analysis of the latest industry trends in each of the sub-segments from 2017 to 2030. The global Military Aircraft MRO market report is segmented based on Aircraft Type, Customer Type, Contract Value Band, Supply Chain Tier, Platform Age, and Region:

- **Aircraft Type Outlook**
  - Fighter Jets
  - Transport Aircraft
  - Attack Helicopters
  - Transport / Utility Helicopters
  - Bombers
  - Special Mission Aircraft
- **Customer Type Outlook**
  - U.S. Air Force
  - Navy / Marine Corps
  - Army Aviation, Europe (NATO Allies)
  - Asia Pacific Allies
  - Middle East FMS (Foreign Military Sales)
  - Latin America & ROW (Rest of World)
  - Government Depots In-House
  - Commercial Contractors
- **Contract Value Band Outlook**
  - Mega (> $1 Billion)
  - Large ($500 Million - $1 Billion)
  - Medium ($100 Million - $500 Million)
  - Small (< $100 Million)
- **Supply Chain Tier Outlook**
  - Tier 1 (Prime Contractors)
  - Tier 2 (Major Subcontractors)
  - Tier 3 (Component Suppliers)
  - Tier 4 (Material / Part Suppliers)
- **Platform Age Outlook**
  - 50+ Years Old
  - 30 - 49 Years Old
  - 20 - 29 Years Old
  - 10 - 19 Years Old
  - < 10 Years Old
- **Regional Outlook**
  - North America
    - U.S.
    - Canada
    - Mexico
  - Europe

        - UK
        - Germany
        - Netherlands
        - Nordics (Sweden, Norway, Denmark)
        - France, Spain, Italy
    - Asia Pacific
        - China
        - Japan
        - India
        - Singapore
        - Australia
        - South Korea
    - Middle East
        - UAE (Dubai)
        - Saudi Arabia (KSA)
        - Israel
    - Africa
        - South Africa
        - Egypt
        - Algeria
    - Latin America
        - Brazil
        - Argentina
        - Colombia

FAQ's- Frequently Asked Questions About This Report.

1. WHAT IS THE MILITARY AIRCRAFT MODERNIZATION MARKET?

The Military Aircraft Modernization Market refers to the global industry focused on upgrading existing military aircraft with advanced avionics, radar, communication, electronic warfare, and weapon systems to enhance operational performance and extend service life.

---

2. WHAT IS THE PROJECTED MARKET SIZE AND GROWTH RATE OF THE MILITARY AIRCRAFT MODERNIZATION MARKET FROM 2025 TO 2030?

The Military Aircraft Modernization Market is projected to witness steady growth between 2025 and 2030, driven by rising defense budgets, fleet life-extension programs, and increasing demand for next-generation combat capabilities.

---

3. WHAT ARE THE KEY TRENDS IN THE MILITARY AIRCRAFT MODERNIZATION MARKET?

Key trends include AI-enabled avionics upgrades, advanced radar and sensor integration, electronic warfare enhancements, digital cockpit retrofits, and adoption of predictive maintenance technologies.

4. WHICH REGIONS DOMINATE THE GLOBAL MILITARY AIRCRAFT MODERNIZATION MARKET?

North America leads the market due to high U.S. defense spending, while Asia-Pacific and Europe are experiencing rapid growth driven by military fleet upgrades and geopolitical security concerns.

5. WHAT FACTORS ARE DRIVING GROWTH IN THE MILITARY AIRCRAFT MODERNIZATION MARKET?

Market growth is driven by aging military aircraft fleets, increasing geopolitical tensions, demand for enhanced mission readiness, technological advancements, and government modernization initiatives.

WHAT THIS REPORT COVERS:

- Market intelligence to support informed business decisions.
- Baseline assessments and forecasts from 2024-2031.
- Identification of opportunities and emerging regulatory trends.
- Entity-level and sector-wide performance benchmarking.
- Competitive positioning and peer benchmarking analysis.
- A structured roadmap to keep you ahead of evolving disclosure requirements.
- Deliverables available in PDF, Excel, PPT, Quadrants, and online Dashboard Versions.

TABLE OF CONTENTS - MILITARY AIRCRAFT MODERNISATION MARKET

Executive Summary.....1
- Market Overview and Key Findings.....1
- Market Size and Growth Projections.....2
- Key Trends and Drivers.....3
- Competitive Landscape Summary.....4

1. Introduction.....5
   1.1. Report Scope and Objectives.....5
   1.2. Research Methodology.....6
   1.3. Key Definitions and Assumptions.....6
   1.4. Market Taxonomy.....7

2. Research Methodology.....8
   2.1. Research Scope.....8
   2.2. Research Methodology.....8
      2.2.1. Market Research Process.....8
      2.2.2. Secondary Research.....9
      2.2.3. Primary Research.....9
      2.2.4. Models for Estimation.....9
   2.3. Market Size Estimation.....10

3. Defense Aviation Market Context.....11
   3.1. Overview of Global Military Aviation Market.....11
   3.2. Role of Aircraft Modernisation within Defense Spending.....12
   3.3. Modernisation vs New Aircraft Procurement Dynamics.....13
   3.4. Regulatory, Export Control, and Procurement Framework.....14

4. Military Aircraft Modernisation Overview.....15
   4.1. Market Definition and Scope.....15
   4.2. Modernisation Categories and Upgrade Types.....16
   4.3. Market Size and Historical Analysis (2019–2024).....17
   4.4. Market Forecast and Projections (2024–2030).....18
   4.5. Market Dynamics.....19
      4.5.1. Drivers.....19
      4.5.2. Restraints.....20
      4.5.3. Opportunities.....20
      4.5.4. Challenges.....21

5. Market Segmentation Analysis.....22

5.1. By Aircraft Type.....22
5.1.1. Fighter jets.....22
5.1.2. Transport Aircraft.....23
5.1.3. Attack Helicopters.....23
5.1.4. Transport/Utility Helicopters.....24
5.1.5. Bombers.....24
5.1.6. Special Mission Aircraft.....25
- Market Size, Growth, and Trends by Aircraft Type.....25

5.2. Customer Type.....26
5.2.1. U.S Air force.....26
5.2.2. Navy/Marine Corps.....26
5.2.3. Army Aviation, Europe (NATOAllies).....27
5.2.4. Asia Pacific Allies.....27
5.2.5. Middle East FMS.....28
5.2.6. Latin America & ROW.....28
5.2.7. Government Depots In-House.....29
5.2.8. Commercial Contractors.....29
- Market Size, Growth, and Trends by Customer Type.....30

5.3. By Contract Value Band.....31
5.3.1. Mega (>$1B).....31
5.3.2. Large ($500M-$1B).....31
5.3.3. Medium ($100M-$500M).....32
5.3.4. Small (<$100M).....32
- Market Size, Growth, and Trends by Contract Brand Value.....33

5.4. By Supply Chain Tier.....34
5.4.1. Tier 1 (Prime Contractors).....34
5.4.2. Tier 2 (Major Subcontractors).....34
5.4.3. Tier 3 (Component Suppliers).....35
5.4.4. Tier 4 (Material/Part Suppliers).....35
- Market Size, Growth, and Trends by Supply Chain Tier.....36

5.5. By Platform Age.....37
5.5.1. 50+ Years Old.....37
5.5.2. 30-49 Years Old.....37
5.5.3. 20-29 Years Old.....38
5.5.4. 10-19 Years Old.....38
5.5.5. <10 Years Old.....39
- Market Size, Growth, and Trends by Platform Age.....39

6. Regional Analysis.....40
6.1. North America.....40
6.1.1. U.S.....41
6.1.2. Canada.....43
6.1.3. Mexico.....45

6.2. Europe.....47
6.2.1. UK.....48
6.2.2. Germany.....50
6.2.3. Netherlands.....52
6.2.4. Nordics (Sweden, Norway, Denmark).....54
6.2.5. France, Spain, Italy.....56

6.3. Asia Pacific.....58
6.3.1. China.....59
6.3.2. Japan.....61
6.3.3. India.....63
6.3.4. Singapore.....65
6.3.5. Australia.....67
6.3.6. South Korea.....69

6.4. Middle East.....71
6.4.1. UAE (Dubai).....72
6.4.2. Saudi Arabia.....74
6.4.3. Israel.....76

6.5. Africa.....78
6.5.1. South Africa.....79
6.5.2. Egypt.....81
6.5.3. Algeria.....83

6.6. Latin America.....85
6.6.1. Brazil.....86
6.6.2. Argentina.....88
6.6.3. Colombia.....90
- Regional Market Size, Growth Drivers, and Key Characteristics.....92

7. Competitive Landscape.....93
7.1. Market Concentration and Structure.....93
7.2. Key Player Market Share Analysis.....94
7.3. Competitive Strategies and Positioning.....95
7.4. Recent Developments (Product Launches, Partnerships, Expansions).....96

8. Company Profiles.....97
8.1. Lockheed Martin.....97
8.1.1. Company Overview.....97

8.1.2. Product/Service Landscape....98
8.1.3. Financial Overview....99
8.1.4. Recent Developments....100

8.2. Thales Group....101
8.2.1. Company Overview....101
8.2.2. Product/Service Landscape....102
8.2.3. Financial Overview....103
8.2.4. Recent Developments....104

8.3. RTX Corporation....105
8.3.1. Company Overview....105
8.3.2. Product/Service Landscape....106
8.3.3. Financial Overview....107
8.3.4. Recent Developments....108

8.4. General Dynamics....109
8.4.1. Company Overview....109
8.4.2. Product/Service Landscape....110
8.4.3. Financial Overview....111
8.4.4. Recent Developments....112

8.5. BAE Systems....113
8.5.1. Company Overview....113
8.5.2. Product/Service Landscape....114
8.5.3. Financial Overview....115
8.5.4. Recent Developments....116

8.6. Boeing....117
8.6.1. Company Overview....117
8.6.2. Product/Service Landscape....118
8.6.3. Financial Overview....119
8.6.4. Recent Developments....120

8.7. Northrop Grumman....121
8.7.1. Company Overview....121
8.7.2. Product/Service Landscape....122
8.7.3. Financial Overview....123
8.7.4. Recent Developments....124

8.8. L3Harris Tech....125
8.8.1. Company Overview....125
8.8.2. Product/Service Landscape....126
8.8.3. Financial Overview....127
8.8.4. Recent Developments....128

8.9. Leonardo S.p.A....129
8.9.1. Company Overview....129
8.9.2. Product/Service Landscape....130
8.9.3. Financial Overview....131

8.9.4. Recent Developments.....132

8.10. Elbit Systems.....133
8.10.1. Company Overview.....133
8.10.2. Product/Service Landscape.....134
8.10.3. Financial Overview.....135
8.10.4. Recent Developments.....136

9. Technology and Modernisation Trends.....137
9.1. Avionics and Mission System Upgrades.....137
9.2. Radar, Sensors, and Electronic Warfare.....138
9.3. Weapons Integration and Open Architecture.....139
9.4. Stealth and Survivability Enhancements.....140
9.5. Digitalisation and Network-Centric Capabilities.....141

10. Regulatory and Procurement Environment.....142
10.1. Defense Procurement Policies.....142
10.2. Export Controls and ITAR Considerations.....143
10.3. Offset Policies and Localisation Requirements.....144
10.4. Impact on Modernisation Programs.....145

11. Market Opportunities and Future Outlook.....146
11.1. Emerging Upgrade Programs.....146
11.2. Regional Growth Hotspots.....147
11.3. Investment and Contracting Opportunities.....148
11.4. Strategic Recommendations for Stakeholders.....149

12. Challenges and Risk Analysis.....150
12.1. Program Delays and Cost Overruns.....150
12.2. Integration and Certification Risks.....151
12.3. Supply Chain and Vendor Dependencies.....152

12.4. Geopolitical and Budgetary Risks.....153

13. Conclusion and Strategic Insights.....154
13.1. Key Takeaways.....154
13.2. Growth Projections Summary.....155
13.3. Investment Attractiveness.....156
13.4. Future Market Direction.....157

14. Appendices.....158
14.1. Appendix A: Research Methodology Details.....158
14.2. Appendix B: List of Abbreviations.....159
14.3. Appendix C: Related Standards and Specifications.....160
14.4. Appendix D: Additional Statistical Tables.....161
14.5. Appendix E: Directory of Key Industry Associations.....162

List of Tables

1. Military Aircraft Modernisation Market Size (2025-2031 USD).....18
2. Military Aircraft Modernisation Market Context (2025-2031 USD).....21
3. Military Aircraft Modernisation Market by region (2025-2031 USD).....34
4. Military Aircraft Modernisation Market NA (2025-2031 USD).....35
5. Military Aircraft Modernisation Market by US region (2025-2031 USD).....36
6. Military Aircraft Modernisation Market by US North South East West Region (2025-2031 USD).....37
7. Military Aircraft Modernisation Market by Canada region (2025-2031 USD).....40
8. Military Aircraft Modernisation Market by Canada North South East West Region (2025-2031 USD).....41
9. Military Aircraft Modernisation Market by Mexico region (2025-2031 USD).....44
10. Military Aircraft Modernisation Market by Mexico North South East West Region (2025-2031 USD).....45
11. Military Aircraft Modernisation Market APAC (2025-2031 USD).....48
12. Military Aircraft Modernisation Market by India region (2025-2031 USD).....50
13. Military Aircraft Modernisation Market by India North South East West Region (2025-2031 USD).....51
14. Military Aircraft Modernisation Market by China region (2025-2031 USD).....54
15. Military Aircraft Modernisation Market by China North South East West Region (2025-2031 USD).....55
16. Military Aircraft Modernisation Market by Japan region (2025-2031 USD).....58
17. Military Aircraft Modernisation Market by Japan North South East West Region (2025-2031 USD).....59
18. Military Aircraft Modernisation Market by South Korea region (2025-2031 USD).....62
19. Military Aircraft Modernisation Market by South Korea North South East West Region (2025-2031 USD).....63
20. Military Aircraft Modernisation Market by Australia region (2025-2031 USD).....66
21. Military Aircraft Modernisation Market by Australia North South East West Region (2025-2031 USD).....67
22. Military Aircraft Modernisation Market by Singapore region (2025-2031 USD).....70
23. Military Aircraft Modernisation Market by Singapore North South East West Region (2025-2031 USD).....71
24. Military Aircraft Modernisation Market Europe (2025-2031 USD).....74
25. Military Aircraft Modernisation Market by UK region (2025-2031 USD).....76
26. Military Aircraft Modernisation Market by UK North South East West Region (2025-2031 USD).....77
27. Military Aircraft Modernisation Market by Netherlands region (2025-2031 USD).....80

28. Military Aircraft Modernisation Market by Netherlands North South East West Region (2025-2031 USD).....81
29. Military Aircraft Modernisation Market by Germany region (2025-2031 USD).....84
30. Military Aircraft Modernisation Market by Germany North South East West Region (2025-2031 USD).....85
31. Military Aircraft Modernisation Market by France Spain Italy region (2025-2031 USD).....88
32. Military Aircraft Modernisation Market by France Spain Italy North South East West Region (2025-2031 USD).....89
33. Military Aircraft Modernisation Market by Nordics region (2025-2031 USD).....92
34. Military Aircraft Modernisation Market by Nordics North South East West Region (2025-2031 USD).....93
35. Military Aircraft Modernisation Market by Middle East (2025-2031 USD).....96
36. Military Aircraft Modernisation Market by UAE region (2025-2031 USD).....98
37. Military Aircraft Modernisation Market by UAE North South East West Region (2025-2031 USD).....99
38. Military Aircraft Modernisation Market by Saudi Arabia region (2025-2031 USD).....102
39. Military Aircraft Modernisation Market by Saudi Arabia North South East West Region (2025-2031 USD).....103
40. Military Aircraft Modernisation Market by Israel region (2025-2031 USD).....104
41. Military Aircraft Modernisation Market by Israel North South East West Region (2025-2031 USD).....104
42. Military Aircraft Modernisation Market Africa (2025-2031 USD).....105
43. Military Aircraft Modernisation Market by South Africa region (2025-2031 USD).....106
44. Military Aircraft Modernisation Market by South Africa North South East West Region (2025-2031 USD).....107
45. Military Aircraft Modernisation Market by Eqypt region (2025-2031 USD).....108
46. Military Aircraft Modernisation Market Eqypt North South East West Region (2025-2031 USD).....108
47. Military Aircraft Modernisation Market Algeria region (2025-2031 USD).....109
48. Military Aircraft Modernisation Market Algeria North South East West Region (2025-2031 USD).....110
49. Military Aircraft Modernisation Market Latin America (2025-2031 USD).....111
50. Military Aircraft Modernisation Market by Brazil region (2025-2031 USD).....112
51. Military Aircraft Modernisation Market by Brazil North South East West Region (2025-2031 USD).....113
52. Military Aircraft Modernisation Market by Chile region (2025-2031 USD).....116
53. Military Aircraft Modernisation Market by Chile North South East West Region (2025-2031 USD).....117
54. Military Aircraft Modernisation Market by Colombia region (2025-2031 USD).....120
55. Military Aircraft Modernisation Market by Colombia North South East West Region (2025-2031 USD).....121
56. Military Aircraft Modernisation Market by Aircraft Type (2025-2031 USD).....122
57. Military Aircraft Modernisation Market by Customer Type (2025-2031 USD).....124

58. Military Aircraft Modernisation Market by Contract Value Band (2025-2031 USD).....126
59. Military Aircraft Modernisation Market by Supply Chain Tier (2025-2031 USD).....128
60. Military Aircraft Modernisation Market by Platform Age (2025-2031 USD).....130
61. Military Aircraft Modernisation Market by Company Overview (2025-2031 USD).....131
62. Military Aircraft Modernisation Market by Lockheed Martin (2025-2031 USD).....132
63. Military Aircraft Modernisation Market by Thales Group (2025-2031 USD).....133
64. Military Aircraft Modernisation Market by RTX Corporation (2025-2031 USD).....134
65. Military Aircraft Modernisation Market by General Dynamics (2025-2031 USD).....138
66. Military Aircraft Modernisation Market by BAE Systems (2025-2031 USD).....142
67. Military Aircraft Modernisation Market by Boeing (2025-2031 USD).....146
68. Military Aircraft Modernisation Market by Northrop Grumman (2025-2031 USD).....150
69. Military Aircraft Modernisation Market by L3Harris Tech (2025-2031 USD).....154
70. Military Aircraft Modernisation Market by Leonardo S.p.A (2025-2031 USD).....158
71. Military Aircraft Modernisation Market by Elbit Systems (2025-2031 USD).....162
72. Military Aircraft Modernisation Market by Standards and Framework 1 (2025-2031 USD).....169
73. Military Aircraft Modernisation Market by Standards and Framework 2 (2025-2031 USD).....171
74. Military Aircraft Modernisation Market by Standards and Framework 3 (2025-2031 USD).....173
75. Military Aircraft Modernisation Market by Standards and Framework 4 (2025-2031 USD).....175
76. Military Aircraft Modernisation Market by Standards and Framework 5 (2025-2031 USD).....177
77. Military Aircraft Modernisation Market by Macro Economic factors 1 (2025-2031 USD).....179
78. Military Aircraft Modernisation Market by Macro Economic factors 2 (2025-2031 USD).....181
79. Military Aircraft Modernisation Market by Macro Economic factors 3 (2025-2031 USD).....183
80. Military Aircraft Modernisation Market by Macro Economic factors 4 (2025-2031 USD).....185
81. Military Aircraft Modernisation Market by Macro Economic factors 5 (2025-2031 USD).....187
82. Military Aircraft Modernisation Market by Market opportunities and Future Outlook (2025-2031 USD).....189
83. Military Aircraft Modernisation Market by Challenges and Risk Analysis (2025-2031 USD).....192
84. Military Aircraft Modernisation Market by Strategic Insights (2025-2031 USD).....195
85. Military Aircraft Modernisation Market by Extra (2025-2031 USD).....200
86. Military Aircraft Modernisation Market by Extra (2025-2031 USD).....200

List of Figures

1. Military Aircraft Modernisation Market Size (2025-2031 USD).....8
2. Military Aircraft Modernisation Market Top Down (2025-2031 USD).....9
3. Military Aircraft Modernisation Market Bottom up (2025-2031 USD).....9
4. Military Aircraft Modernisation Market Primary Research (2025-2031 USD).....9
5. Military Aircraft Modernisation Market Secondary research (2025-2031 USD).....9
6. Military Aircraft Modernisation Market Context (2025-2031 USD).....21
7. Military Aircraft Modernisation Market DROC (2025-2031 USD).....21
8. Military Aircraft Modernisation Market by Region (2025-2031 USD).....34
9. Military Aircraft Modernisation Market North America (2025-2031 USD).....35
10. Military Aircraft Modernisation Market US region (2025-2031 USD).....36
11. Military Aircraft Modernisation Market US North South East West Region (2025-2031 USD).....38
12. Military Aircraft Modernisation Market Canada region (2025-2031 USD).....40
13. Military Aircraft Modernisation Market Canada North South East West Region (2025-2031 USD).....42
14. Military Aircraft Modernisation Market Mexico region (2025-2031 USD).....44
15. Military Aircraft Modernisation Market Mexico North South East West Region (2025-2031 USD).....46
16. Military Aircraft Modernisation Market APAC (2025-2031 USD).....48
17. Military Aircraft Modernisation Market India region (2025-2031 USD).....50
18. Military Aircraft Modernisation Market India North South East West Region (2025-2031 USD).....52
19. Military Aircraft Modernisation Market China region (2025-2031 USD).....54
20. Military Aircraft Modernisation Market China North South East West Region (2025-2031 USD).....56
21. Military Aircraft Modernisation Market Japan region (2025-2031 USD).....58
22. Military Aircraft Modernisation Market Japan North South East West Region (2025-2031 USD).....60
23. Military Aircraft Modernisation Market South Korea region (2025-2031 USD).....62
24. Military Aircraft Modernisation Market South Korea North South East West Region (2025-2031 USD).....64
25. Military Aircraft Modernisation Market Australia region (2025-2031 USD).....66
26. Military Aircraft Modernisation Market Australia North South East West Region (2025-2031 USD).....68
27. Military Aircraft Modernisation Market Singapore region (2025-2031 USD).....70
28. Military Aircraft Modernisation Market Singapore North South East West Region (2025-2031 USD).....72

29. Military Aircraft Modernisation Market Europe region (2025-2031 USD).....74
30. Military Aircraft Modernisation Market UK region (2025-2031 USD).....76
31. Military Aircraft Modernisation Market UK North South East West Region (2025-2031 USD).....78
32. Military Aircraft Modernisation Market Netherlands region (2025-2031 USD).....80
33. Military Aircraft Modernisation Market Netherlands North South East West Region (2025-2031 USD).....82
34. Military Aircraft Modernisation Market Germany region (2025-2031 USD).....84
35. Military Aircraft Modernisation Market Germany North South East West Region (2025-2031 USD).....86
36. Military Aircraft Modernisation Market France Spain Italy region (2025-2031 USD).....88
37. Military Aircraft Modernisation Market France Spain Italy North South East West Region (2025-2031 USD).....90
38. Military Aircraft Modernisation Market Nordics region (2025-2031 USD).....92
39. Military Aircraft Modernisation Market Nordics North South East West Region (2025-2031 USD).....94
40. Military Aircraft Modernisation Market Middle East (2025-2031 USD).....96
41. Military Aircraft Modernisation Market UAE region (2025-2031 USD).....98
42. Military Aircraft Modernisation Market UAE North South East West Region (2025-2031 USD).....100
43. Military Aircraft Modernisation Market by Saudi Arabia region (2025-2031 USD).....102
44. Military Aircraft Modernisation Market by Saudi Arabia North South East West Region (2025-2031 USD).....104
45. Military Aircraft Modernisation Market by Israel region (2025-2031 USD).....104
46. Military Aircraft Modernisation Market Israel North South East West Region (2025-2031 USD).....105
47. Military Aircraft Modernisation Market Africa (2025-2031 USD).....106
48. Military Aircraft Modernisation Market by South Africa region (2025-2031 USD).....107
49. Military Aircraft Modernisation Market by South Africa North South East West Region (2025-2031 USD).....108
50. Military Aircraft Modernisation Market by Eqypt region (2025-2031 USD).....108
51. Military Aircraft Modernisation Market Eqypt North South East West Region (2025-2031 USD).....108
52. Military Aircraft Modernisation Market Algeria region (2025-2031 USD).....109
53. Military Aircraft Modernisation Market Algeria North South East West Region (2025-2031 USD).....110
54. Military Aircraft Modernisation Market Latin America (2025-2031 USD).....111
55. Military Aircraft Modernisation Market by Brazil region (2025-2031 USD).....112
56. Military Aircraft Modernisation Market by Brazil North South East West Region (2025-2031 USD).....114
57. Military Aircraft Modernisation Market by Chile region (2025-2031 USD).....116
58. Military Aircraft Modernisation Market by Chile North South East West Region (2025-2031 USD).....118

59. Military Aircraft Modernisation Market by Colombia region (2025-2031 USD).....120
60. Military Aircraft Modernisation Market by Colombia North South East West Region (2025-2031 USD).....122
61. Military Aircraft Modernisation Market by Aircraft Type(2025-2031 USD).....123
62. Military Aircraft Modernisation Market by Customer Type (2025-2031 USD).....125
63. Military Aircraft Modernisation Market by Contract Value Band (2025-2031 USD).....127
64. Military Aircraft Modernisation Market by Supply Chain Tier (2025-2031 USD).....129
65. Military Aircraft Modernisation Market by Platform Age (2025-2031 USD).....131
66. Military Aircraft Modernisation Market by Company Overview (2025-2031 USD).....133
67. Military Aircraft Modernisation Market by Lockheed Martin (2025-2031 USD).....135
68. Military Aircraft Modernisation Market by Company segment (2025-2031 USD).....136
69. Military Aircraft Modernisation Market by Thales Group (2025-2031 USD).....138
70. Military Aircraft Modernisation Market by Company segment (2025-2031 USD).....139
71. Military Aircraft Modernisation Market by RTX Corporation (2025-2031 USD).....140
72. Military Aircraft Modernisation Market by Company segment (2025-2031 USD).....141
73. Military Aircraft Modernisation Market by General Dynamics (2025-2031 USD).....142
74. Military Aircraft Modernisation Market by Company segment (2025-2031 USD).....143
75. Military Aircraft Modernisation Market by BAE Systems (2025-2031 USD).....143
76. Military Aircraft Modernisation Market by Company segment (2025-2031 USD).....144
77. Military Aircraft Modernisation Market by Boeing (2025-2031 USD).....147
78. Military Aircraft Modernisation Market by Company segment (2025-2031 USD).....148
79. Military Aircraft Modernisation Market by Northrop Grumman (2025-2031 USD).....151
80. Military Aircraft Modernisation Market by Company segment (2025-2031 USD).....152
81. Military Aircraft Modernisation Market by L3Harris Tech (2025-2031 USD).....155
82. Military Aircraft Modernisation Market by Company segment (2025-2031 USD).....156
83. Military Aircraft Modernisation Market by Leonardo S.p.A. (2025-2031 USD).....159
84. Military Aircraft Modernisation Market by Company segment (2025-2031 USD).....160
85. Military Aircraft Modernisation Market by Elbit Systems (2025-2031 USD).....163
86. Military Aircraft Modernisation Market by Company segment (2025-2031 USD).....164
87. Military Aircraft Modernisation Market by Standards and Framework 1 (2025-2031 USD).....170
88. Military Aircraft Modernisation Market by Standards and Framework 2 (2025-2031 USD).....172

89. Military Aircraft Modernisation Market by Standards and Framework 3 (2025-2031 USD).....174
90. Military Aircraft Modernisation Market by Standards and Framework 4 (2025-2031 USD).....176
91. Military Aircraft Modernisation Market by Standards and Framework 5 (2025-2031 USD).....178
92. Military Aircraft Modernisation Market by Macro Economic factors 1 (2025-2031 USD).....180
93. Military Aircraft Modernisation Market by Macro Economic factors 2 (2025-2031 USD).....182
94. Military Aircraft Modernisation Market by Macro Economic factors 3 (2025-2031 USD).....184
95. Military Aircraft Modernisation Market by Macro Economic factors 4 (2025-2031 USD).....186
96. Military Aircraft Modernisation Market by Macro Economic factors 5 (2025-2031 USD).....188
97. Military Aircraft Modernisation Market by Market opportunities and Future Outlook (2025-2031 USD).....190
98. Military Aircraft Modernisation Market by Challenges and Risk Analysis (2025-2031 USD).....193
99. Military Aircraft Modernisation Market by Strategic Insights (2025-2031 USD).....196
100. Military Aircraft Modernisation Market by Extra (2025-2031 USD).....200
101. Military Aircraft Modernisation Market by Extra (2025-2031 USD).....200